



Endoscopic submucosal dissection versus endoscopic mucosal resection for colorectal laterally spreading tumors (> 20 mm): a systematic review and meta-analysis

Q. Zeng^{1,2} · L. Zhang^{1,2} · T. Zhang^{1,2} · T. Kong^{1,2} · S. Xu^{1,2} · C. Lan^{1,2} · B. He^{1,2}

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Abstract

Background Colorectal laterally spreading tumors (LSTs) larger than 20 mm are significant precursors to colorectal cancer. Endoscopic submucosal dissection (ESD) and endoscopic mucosal resection (EMR) are widely used treatments, but their comparative efficacy and safety remain controversial. This systematic review and meta-analysis aimed to compare ESD and EMR in terms of local recurrence, perforation, and delayed bleeding for colorectal LSTs > 20 mm.

Methods Following PRISMA guidelines, a comprehensive literature search was conducted in PubMed, Embase, and Cochrane databases from inception to 1 May 2025. Studies comparing ESD and EMR for LSTs > 20 mm were included. Data were extracted and quality assessed using the Newcastle–Ottawa Scale and Cochrane Risk of Bias Tool. Random-effects models were used to calculate pooled odds ratios (ORs) with 95% confidence intervals (CIs).

Results Eight studies comprising 3024 patients were included. ESD was associated with significantly lower local recurrence compared to EMR (0.62% versus 12.02%; OR 0.07, 95% CI 0.03–0.13, $p < 0.01$). However, ESD had a higher risk of perforation (5.38% versus 1.53%; OR 3.16, 95% CI 1.89–5.29, $p < 0.01$). No significant difference was found in delayed bleeding rates (3.65% versus 3.75%; OR 1.01, 95% CI 0.65–1.55, $p = 0.98$). Heterogeneity was negligible across all outcomes ($I^2 < 5\%$).

Conclusions ESD significantly reduces recurrence risk for colorectal LSTs > 20 mm compared with EMR but carries a higher perforation risk. Bleeding rates are comparable between the two techniques. ESD is recommended in expert centers where technical proficiency can mitigate perforation risk, while EMR-treated patients require closer surveillance due to higher recurrence. Future studies should focus on optimizing techniques and clarifying strategies for different LST subtypes and locations.

Keywords Neoplasms · Endoscopic mucosal resection · Endoscopic submucosal dissection · Recurrence · Complications

Introduction

In 2020, colorectal cancer (CRC) accounted for over 1.9 million new cases and approximately 0.9 million deaths globally, making it the third leading cause of cancer-related mortality worldwide [1]. The incidence of CRC has continued

to rise in recent years, particularly in developing countries. This high incidence and mortality underscore the critical importance of early detection and intervention. Laterally spreading tumors (LSTs) have been recognized as important precursors of CRC [2].

✉ T. Zhang
305514271@qq.com
Q. Zeng
2027863081@qq.com
L. Zhang
286535795@qq.com
T. Kong
424201342@qq.com
S. Xu
2026682455@qq.com

C. Lan
369911243@qq.com
B. He
hbb1987@qq.com

¹ Department of Gastroenterology, The Affiliated Hospital of Southwest Medical University, Nanchong Central Hospital, Luzhou 646000, Sichuan, China

² Nanchong Central Hospital, Nanchong 637000, China

LSTs are defined as flat, broad-based lesions with a minimum diameter of 10 mm, predominantly located in the colorectum. Based on endoscopic morphology, LSTs are classified into two subtypes: granular type (LST-G) and non-granular type (LST-NG) [3]. These lesions have the potential to progress to high-grade intraepithelial neoplasia, with incidence rates reported up to 33.8% [4, 5]. Therefore, the early identification and resection of colorectal LSTs are essential for the prevention and treatment of carcinogenesis.

Compared with surgical intervention, endoscopic treatment has been widely adopted owing to its minimally invasive nature, rapid recovery, and cost-effectiveness. The main endoscopic techniques for LSTs include endoscopic submucosal dissection (ESD) and endoscopic mucosal resection (EMR). For lesions larger than 20 mm, EMR often necessitates piecemeal resection (PEMR), whereas ESD enables en bloc resection [6]. Several studies have compared the efficacy and safety of ESD and EMR for large colorectal LSTs. However, for LSTs exceeding 20 mm in diameter, the optimal choice between ESD and EMR remains controversial [6–9]. To date, no systematic review or meta-analysis has specifically compared the efficacy and safety of ESD versus EMR for the resection of colorectal LSTs greater than 20 mm. Therefore, this systematic review aims to comprehensively evaluate the efficacy and safety of ESD and EMR for colorectal LSTs larger than 20 mm, providing evidence-based guidance for clinical decision-making.

Materials and methods

The reporting of this systematic review follows the recommendations of the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (Supplementary Material 1) [10].

Search strategy:

An electronic literature search was conducted by an experienced librarian with supervision from the study's principal investigator. Multiple databases were searched from inception to 1 May 2025, to identify eligible studies. The databases searched included PubMed, Embase, and Cochrane. The search strategies employed in PubMed and Cochrane Library databases were as follows: (((colorectal tumors) OR (laterally spreading tumor)) AND (endoscopic submucosal dissection) OR (ESD)) AND ((piecemeal endoscopic mucosal resection) OR (pEMR) OR (piecemeal EMR) OR (pmEMR) OR (EMR-precutting) OR (EMR-P) OR (endoscopic piecemeal mucosal resection) OR (EPMR)). The search strategy applied in the Embase database was: (colorectal AND ("tumors"/exp OR tumors) OR laterally) AND spreading AND ("tumor"/exp OR tumor) AND (endoscopic AND submucosal AND ("dissection"/exp OR dissection) OR esd) AND (((piecemeal AND endoscopic AND mucosal

AND ("resection"/exp OR resection) OR pembr OR piecemeal) AND emr OR pmembr OR "emr precutting" OR "emr p" OR endoscopic) AND piecemeal AND mucosal AND ("resection"/exp OR resection) OR epmr).

Study selection:

Inclusion criteria: (1) Studies involving LSTs with lesions measuring > 20 mm; (2) Comparative studies reporting primary and secondary outcomes of ESD versus EMR.

Exclusion criteria: (1) Studies including LSTs < 20 mm or other lesions preventing outcome extraction; (2) Reviews, meta-analyses, letters, editorials; (3) Case reports/series or studies with < 30 patients per ESD/EMR group; (4) Animal studies.

Two reviewers (initials, Q.Z. and L.Z.) independently screened the titles and abstracts of all records identified by the search. Full-text articles of potentially eligible studies were then independently assessed by the same two reviewers for eligibility. Disagreements at any stage were resolved through discussion or, if necessary, by consulting a third reviewer (T.Z.). The reasons for excluding studies at the full-text stage were recorded. The selection process was documented using a PRISMA flow diagram (Fig. 1).

Data extraction:

Data were extracted independently by two reviewers (Q.Z. and T.K.) using a predefined data extraction form. The extracted data included: first author, publication year, study design (single-center/multicenter), total patients, EMR/ESD techniques, recurrence, and adverse events (delayed bleeding, perforation). Any discrepancies in data extraction were resolved by consensus or by involving a third reviewer (S.X.). If necessary, study authors were contacted for clarification or additional data. The data extraction forms and analytic code are available from the corresponding author upon reasonable request.

Quality assessment:

The risk of bias was assessed independently by two reviewers (Q.Z. and T.K.) using the Cochrane Risk of Bias Tool (RoB 2) for randomized trials and the Newcastle–Ottawa Scale (NOS) for nonrandomized controlled studies. Disagreements were resolved by discussion or by consulting a third reviewer (T.Z.).

Outcomes and definitions

Primary outcome: Local recurrence (defined as biopsy-positive recurrence from resection specimens, regardless of endoscopic appearance).

Secondary outcomes: Perforation (endoscopic evidence of mural defect or radiologically confirmed free air on computed tomography (CT)/plain abdominal films). Delayed bleeding (postprocedural hemorrhage at resection site requiring hemostasis).

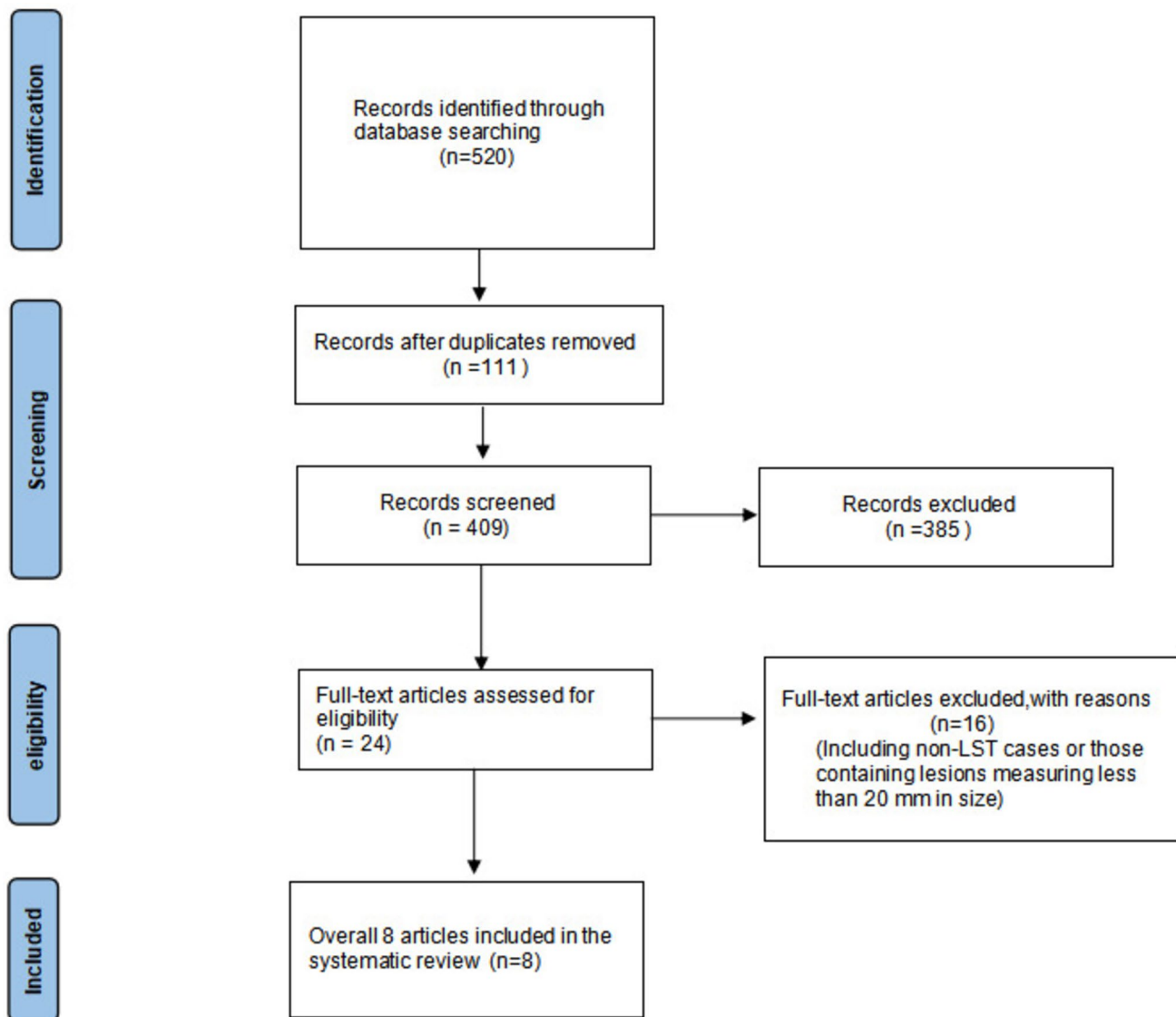


Fig. 1 Flow chart of the systematic review

Statistical analysis

Statistical analyses were performed using the Cochrane Collaboration's RevMan 5.4 software. Statistical analyses were performed using the Cochrane Collaboration's RevMan 5.4 software. Data were analyzed using random-effects models, with forest plots generated for each outcome. Odds ratios (ORs) with 95% confidence intervals (CIs) were calculated. Heterogeneity was assessed via Q -test and I^2 statistics. Sensitivity analysis employed the leave-one-out method to evaluate robustness. Publication bias was initially evaluated through funnel plot visualization and confirmed with Egger's test (Stata 18.0). Statistical significance was defined as $p < 0.05$ (heterogeneity threshold: $p < 0.1$).

Results

Search results and population characteristics

A total of 111 duplicate records were excluded using End-Note. After removing reviews, meta-analyses, letters, editorials, case reports/series, and ineligible publications, eight studies were ultimately included (see PRISMA flowchart, Fig. 1).

The analysis comprised 3024 patients: 1451 underwent ESD and 1573 received EMR (including 588 piecemeal EMR procedures). Included studies consisted of one RCT and seven retrospective studies (six single-center, two multi-center), with geographical distribution as follows: three from South Korea, three from Japan, one from Germany, and one

from France. Detailed study characteristics are summarized in Table 1. We used the Newcastle–Ottawa Scale for non-randomized studies to assess methodology quality, which gave a mean score of 7.7 (range 6–9) (Table 2) [18]. For the randomized trials, we used Cochrane’s Risk of Bias assessment (Table 3) [19]. General characteristics of the studies and characteristics of the included lesions were shown in Table 1.

Meta-analysis outcomes

Local recurrence (eight studies, $n = 3024$): Pooled recurrence rates: ESD 0.62% versus EMR 12.02%. OR 0.07 (95% CI 0.03–0.13; $p < 0.01$) with no heterogeneity ($I^2 = 0\%$, random-effects model) (Fig. 2). No publication bias was detected by funnel plot inspection or Egger’s test ($p = 0.78$) (Figs. 3 and 4).

Delayed bleeding (eight studies, $n = 3024$): Pooled rates: ESD 3.65% versus EMR 3.75%. OR 1.01 (95% CI 0.65–1.55; $p = 0.98$) with no heterogeneity ($I^2 = 5\%$, random-effects model) (Fig. 5). No publication bias was observed via funnel plot or Egger’s test ($p = 0.42$) (Figs. 6 and 7).

Perforation (eight studies, $n = 3024$): Pooled rates: ESD 5.38% versus EMR 1.53%. OR 3.16 (95% CI 1.89–5.29; $p < 0.01$) with no heterogeneity ($I^2 = 0\%$, random-effects model) (Fig. 8). No publication bias was detected by funnel plot or Egger’s test ($p = 0.45$) (Figs. 9 and 10).

Sensitivity analysis

We performed a sensitivity analysis by excluding one study at a time to evaluate its impact on the main summary estimate. This process showed that no single study significantly influenced the outcome or the heterogeneity.

Quality of evidence

The GRADE working group approach was used to rate the quality of evidence. Based on the predominance of observational studies and negligible heterogeneity ($I^2 < 5\%$), the overall evidence was rated as low certainty. However, for local recurrence, the certainty was upgraded to moderate due to the very large magnitude of effect (OR 0.06).

Discussion

Endoscopic submucosal dissection (ESD), as a minimally invasive technique, has been widely recognized as an effective approach for the management of superficial colorectal tumors larger than 2 cm. Compared with conventional surgery, ESD offers significant advantages in terms of patient tolerability, faster recovery, and greater cost-effectiveness

Table 1 Main characteristics of the included studies

Author (year)	Study design Origin	Patients	Technique		Recurrence		Bleeding		Perforation	
			ESD	EMR/PEMR	ESD	EMR/PEMR	ESD	EMR/PEMR	ESD	EMR/PEMR
Kouladouros et al. (2024) [11]	Multicenter retrospective Germany	130	68	62	1	16	10	4	0	0
Sudo et al. (2011) [12]	Monocentric retrospective Japan	676	167	509	0	79	2	12	6	7
Saito et al. (2009) [13]	Monocentric retrospective Japan	373	145	228	3	33	2	7	9	3
Terasaki et al. (2011) [14]	Monocentric retrospective Japan	239	61	178	0	14	7	15	0	3
Oh et al. (2021) [15]	Monocentric retrospective Korea	180	90	90	0	0	2	2	6	2
Ham et al. (2019) [16]	Monocentric retrospective Korea	544	429	115	2	8	14	7	21	3
Lee et al. (2011) [17]	Monocentric retrospective Korea	523	314	209	2	31	2	2	25	2
Jacques et al. (2024) [18]	Multicenter RCT France	359	177	182	1	8	14	10	11	4

Table 2 Risk of bias across studies assessed using the Newcastle–Ottawa Scale

Author (year)	Selection (0–4 score)				Comparability (0–2 score)	Outcomes (0–3 score)			Total score for study quality
	Is the case definition adequate	Representativeness of the cases	Selection of controls	Definition of controls	Comparability of cases and controls on the basis of the design or analysis	Ascertainment of exposure	Same method of ascertainment for cases and controls	Non-response rate	
Kouladouros et al. (2024)	1	1	1	1	2	1	1	0	8
Sudo et al. (2011)	1	1	1	1	1	1	0	0	6
Saito et al. (2009)	1	1	1	1	1	1	1	1	8
Terasaki et al. (2011)	1	1	1	1	1	1	1	1	8
Oh et al. (2021)	1	1	1	1	2	1	1	1	9
Ham et al. (2019)	1	1	1	1	1	1	1	0	7
Lee et al. (2011)	1	1	1	1	1	1	1	1	8

[20]. One of the key findings of this meta-analysis is the significant superiority of ESD in reducing recurrence rates of colorectal laterally spreading tumors (LSTs) larger than 20 mm. The overall recurrence rates in the ESD and EMR groups were 0.62% and 12.02%, respectively, which aligns closely with the results of studies such as that by Russo et al. [21]. The higher recurrence rate in the EMR group observed in this meta-analysis is associated with both lesion size and the use of piecemeal resection [14, 22, 23]. This is further supported by the findings of Motomi Terasaki et al., who reported that the local recurrence rate for 40 mm LSTs was significantly higher than that for 20–39 mm lesions ($p < 0.001$), and that lesions resected in three or more pieces had a notably higher recurrence rate compared with those resected in two pieces or en bloc ($p = 0.013$). Similarly, in the study by Yutaka Saito et al., the recurrence rate was approximately 2% in the ESD group compared with approximately 14% in the EMR group, and recurrence after EMR occurred more frequently in piecemeal resection cases. Therefore, when lesions are large and piecemeal resection is anticipated (especially > 40 mm), ESD may be preferred to reduce the risk of recurrence. In contrast, if EMR can achieve en bloc or limited-piece resection, the recurrence risk appears to be relatively controllable. These results highlight the crucial role of resection completeness and lesion size in recurrence risk.

Because all lesions analyzed in this study were larger than 20 mm, en bloc resection is often difficult to achieve using conventional EMR, necessitating widespread adoption of piecemeal resection. This approach has inherent drawbacks:

piecemeal resection significantly increases the risk of incomplete removal, renders precise histological margin assessment (R0 resection) extremely challenging, and makes it difficult to ensure complete eradication of neoplastic tissue. Residual adenomatous tissue has been shown to have a rapid regenerative capacity and may increase the risk of recurrence [24, 25]. The high proportion of piecemeal resections in the EMR group directly explains its significantly elevated recurrence rate. In addition to the technical comparison between ESD and EMR, the use of adjuvant thermal ablation techniques—such as snare tip soft coagulation (STSC) or argon plasma coagulation (APC)—applied to the resection margins following piecemeal EMR has emerged as a promising strategy to reduce local recurrence. A systematic review and meta-analysis by Chandan et al. demonstrated that STSC after EMR for LSTs > 20 mm significantly lowered adenoma recurrence at the first surveillance colonoscopy (pooled rate: 6%; OR 0.27, $p < 0.001$) [26]. Similarly, a recent meta-analysis of randomized trials by Santo et al. confirmed that thermal ablation (including both STSC and APC) of post-EMR margins reduced recurrence rates (RR = 0.31, $p < 0.001$) without increasing the risk of adverse events or delayed bleeding [27]. These findings suggest that margin ablation may serve as a safe and effective adjunct to EMR, particularly in settings where ESD is not readily available or feasible. However, its implementation should be guided by lesion characteristics and endoscopist expertise, and future studies should directly compare EMR with adjuvant ablation versus ESD in large LSTs.

Table 3 Cochrane Collaboration tool to assess risk of bias

Study	Random sequence generation (selection bias)	Allocation concealment (selection bias)	Blinding of participants and personnel (performance bias)	Blinding of outcome assessment (detection bias)	Incomplete outcome data (attrition bias)	Selective reporting (reporting bias)
Jacques et al. (2024)	Low risk	Low risk	High risk	Low risk	High risk	Low risk

Owing to the difficulty in assessing margins after piecemeal resection, this study omitted the evaluation of R0 resection rate, which further underscores the pathological assessment differences between the two techniques and their implications for follow-up management. Based on the current study and literature evidence [13, 23, 28], we recommend that patients who achieve curative en bloc ESD, given their extremely low recurrence risk, should undergo standard surveillance colonoscopy at 1 year postprocedure. In contrast, for cases managed with piecemeal ESD or any EMR, due to the significantly increased risk of residual lesion and recurrence, we strongly recommend shortening the surveillance interval to 6 months to enable early detection and intervention for residual or recurrent lesions.

In terms of safety, the risk of perforation with ESD is higher, a concern that is particularly prominent in Western countries, leading to EMR being more commonly adopted as the mainstream approach [29]. Our findings are consistent with this, showing a higher perforation risk for ESD compared with EMR. Right-sided colonic lesions and longer procedure duration are significant risk factors for ESD-related perforation, while lesion size itself is not directly associated with perforation risk [30, 31]. This suggests that, in the hands of experienced endoscopists, even larger lesions can be safely managed without a proportional increase in perforation risk, which carries important clinical implications. Endoscopists should exercise particular caution during ESD for right-sided LSTs. Technological advancements are crucial for reducing perforation risk; for example, Okamoto et al. demonstrated that the use of novel Mucosectomy knives during ESD significantly shortened procedure time and reduced the risk of perforation [32]. Notably, the widespread adoption of ESD is limited by its steep learning curve. The number of procedures required to achieve proficiency varies considerably based on training pathways. In structured, tutored settings such as those in Japan, trainees may achieve competence after approximately 30–40 colorectal ESDs [33, 34]. However, in Western countries, where an untutored, prevalence-based approach is often adopted, the learning curve is substantially longer, with some studies reporting a requirement of up to 280 procedures to achieve expert-level resection speed and safety [35]. These differences underscore the importance of dedicated training programs, mentorship, and case selection during the initial learning phase to optimize outcomes and minimize complications. This underscores the importance of systematic training and accumulation of surgical experience. Our study also showed that the vast majority of perforations could be managed successfully with endoscopic clipping or conservative treatment, with only a minority requiring surgical intervention, indicating that ESD is controllable and safe in experienced hands.

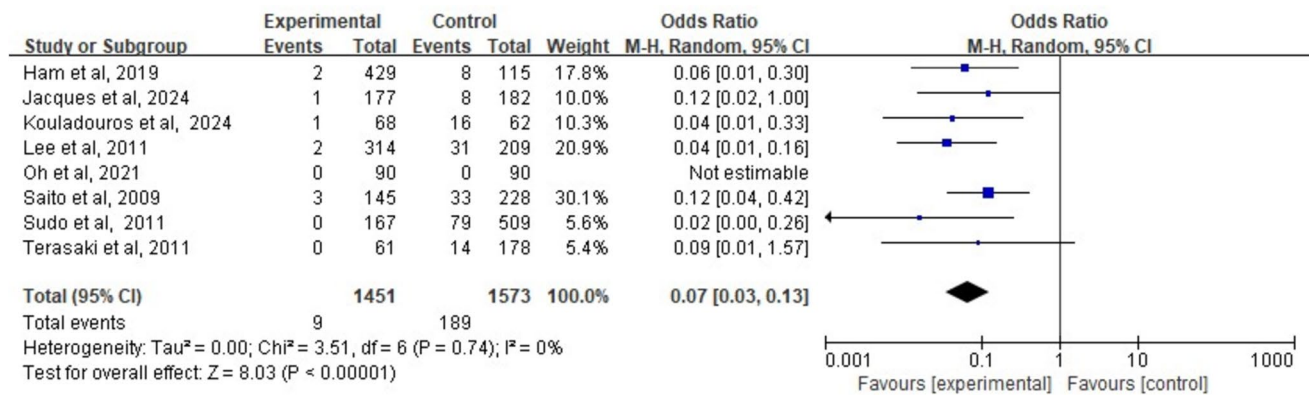
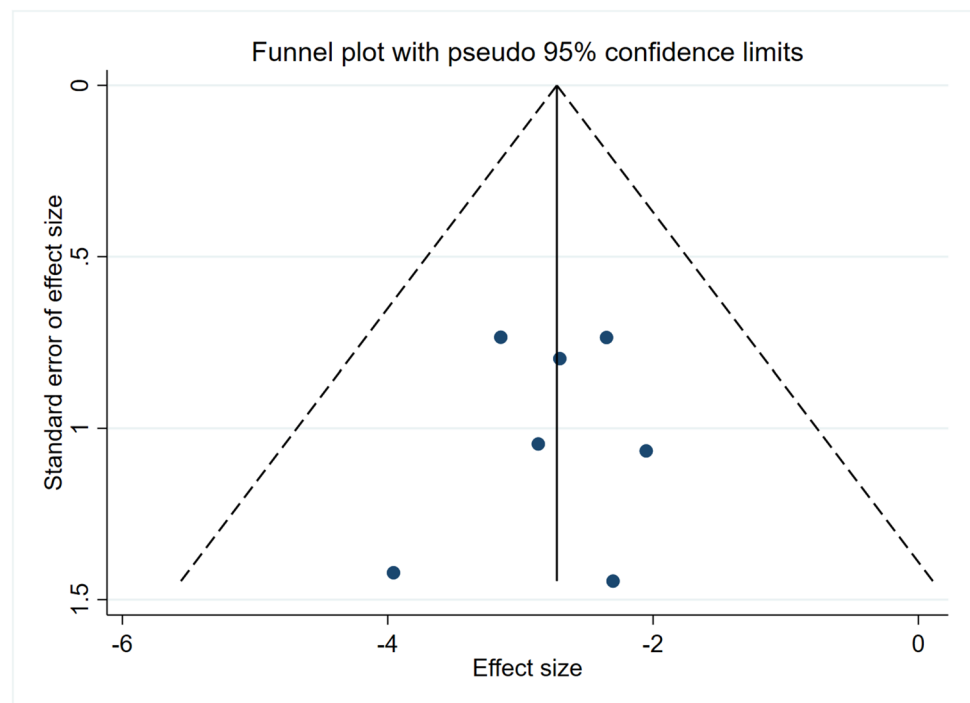


Fig. 2 Forest plot, local recurrence rate

Fig. 3 Evaluation of publication bias for recurrence using funnel plot



No significant difference was observed between ESD and EMR in terms of postprocedural bleeding, with overall bleeding rates of 3.65% and 3.75%, respectively. Rectal location and longer ESD procedure time have been identified as risk factors for post-ESD bleeding [30]. Of note, Motomi Terasaki et al. found that in comparison with conventional ESD, hybrid ESD (lesions were resected by snaring after circumferential incision and limited submucosal dissection), which significantly shortens operative time, was associated with no delayed bleeding events. This suggests that optimizing procedural workflow and reducing operative time may help decrease bleeding complications. The vast majority of bleeding episodes were successfully

managed with endoscopic clipping without the need for transfusion or additional surgery.

Limitations

Most studies included in this meta-analysis were single-center or retrospective in nature. To minimize the risk of bias, we used the Newcastle–Ottawa Scale to assess the quality of non-randomized studies. Furthermore, owing to missing original data, we were unable to compare the potential impact of different LST subtypes or lesion locations (e.g., proximal colon versus distal colon/rectum) on the efficacy and safety of the

Fig. 4 Evaluation of publication bias for recurrence using Egger's test

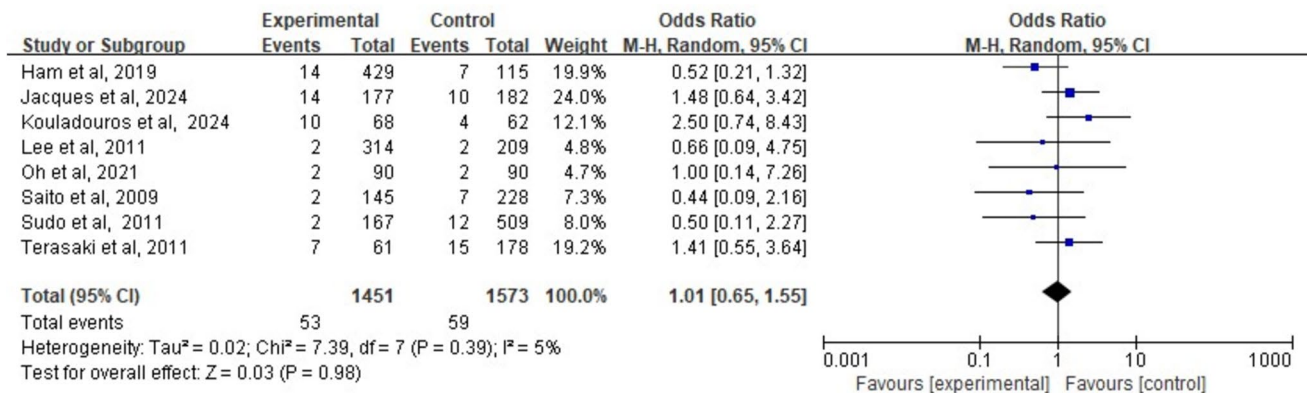
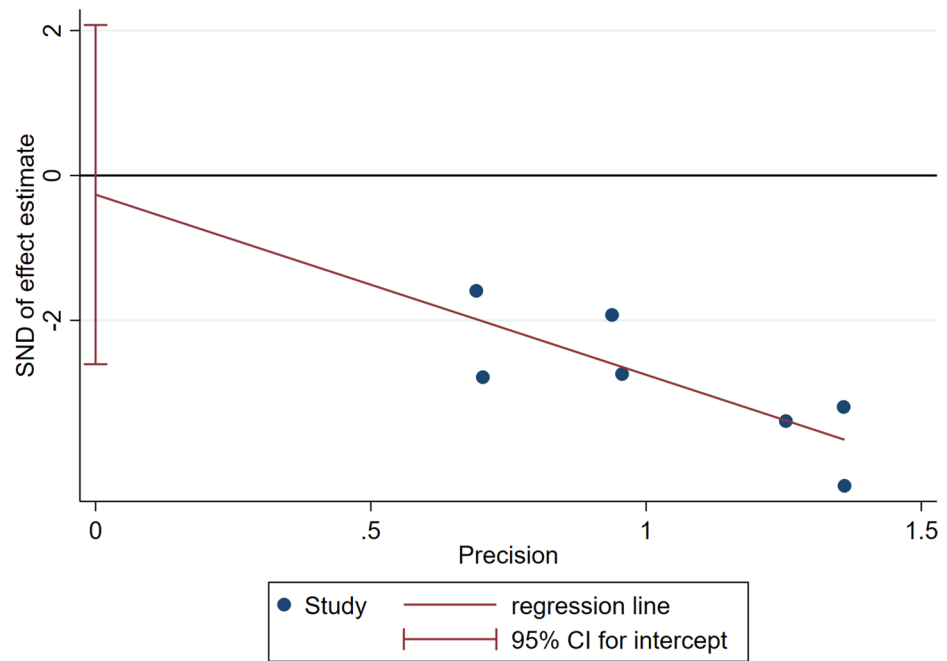


Fig. 5 Forest plot, bleeding rate

two techniques. Since different subtypes and locations may vary in biological behavior, technical difficulty, and prognosis (e.g., more advanced histology such as high-grade intraepithelial neoplasia or intramucosal carcinoma may increase the difficulty of en bloc resection), the absence of this information may affect the interpretation of the results. Future prospective, multicenter studies, especially those collecting detailed lesion characteristics, are needed to more comprehensively evaluate the value of ESD and EMR in different clinical scenarios.

Conclusions

This meta-analysis confirms that, for colorectal LSTs larger than 20 mm, endoscopic submucosal dissection (ESD) offers a clear advantage over endoscopic mucosal resection (EMR) in significantly reducing tumor recurrence rates. Despite the relatively higher risk of perforation and the steep learning curve associated with ESD,

Fig. 6 Evaluation of publication bias for bleeding using funnel plot

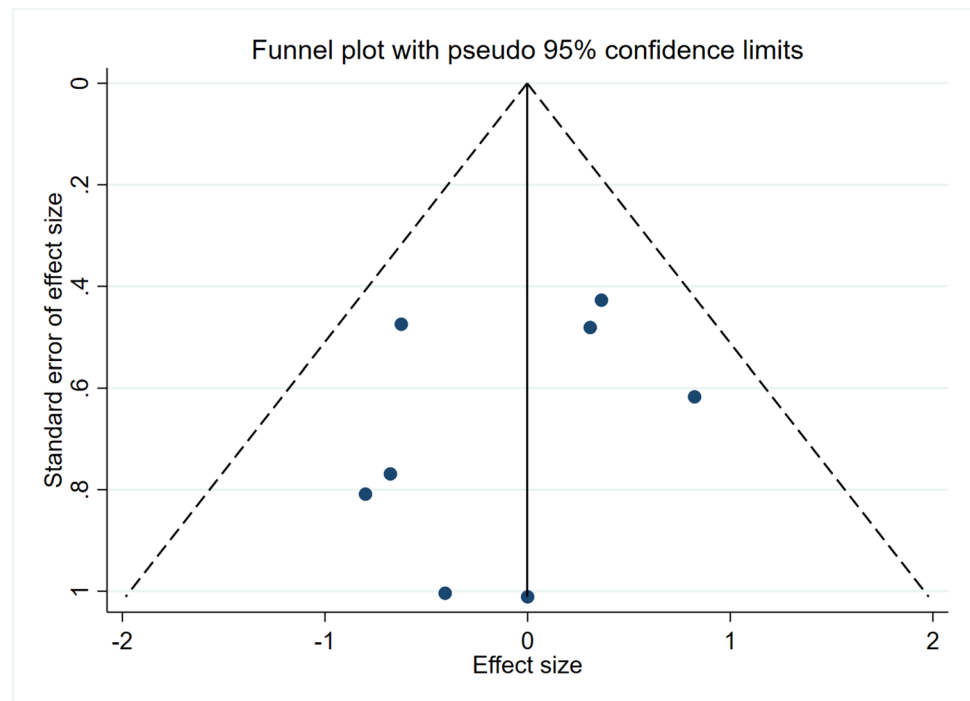
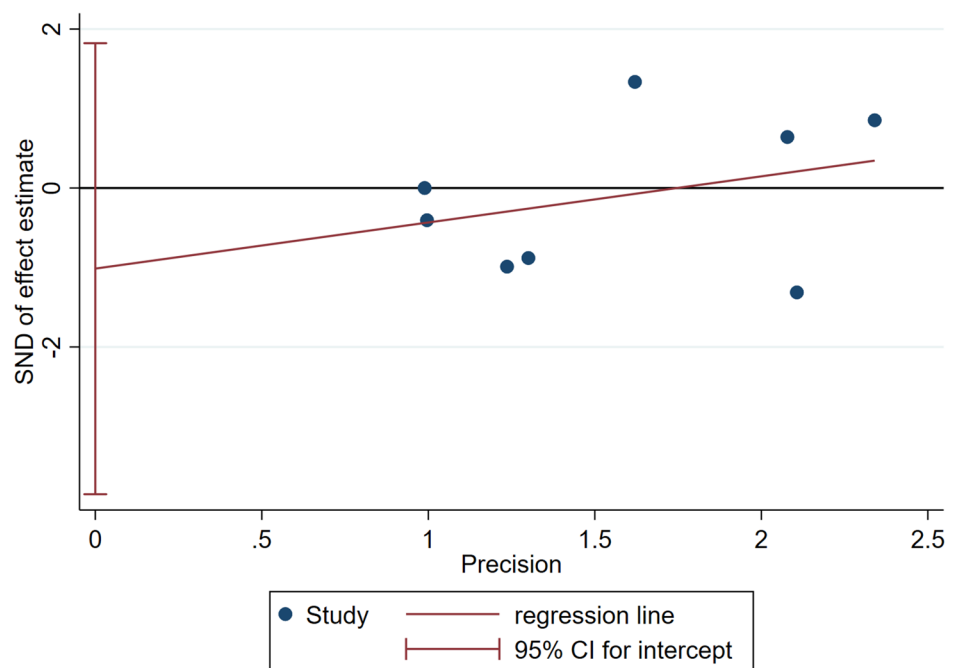


Fig. 7 Evaluation of publication bias for bleeding using Egger's test



evidence suggests that these challenges can be effectively mitigated through technological innovations (such as dedicated devices and hybrid ESD), rigorous operator training and experience accumulation (with emphasis on case volume thresholds), and targeted preventive measures for high-risk factors (e.g., right-sided colon lesions). Most complications are manageable endoscopically. In terms of bleeding risk, both techniques are comparable. Therefore,

in centers with well-established ESD expertise and experience, ESD should be regarded as the preferred method for the treatment of colorectal LSTs larger than 20 mm, providing superior long-term oncological outcomes (extremely low recurrence rates) while maintaining the safety and efficiency of minimally invasive therapy. For centers at an earlier stage of the ESD learning curve or for selected high-risk cases, EMR remains an important

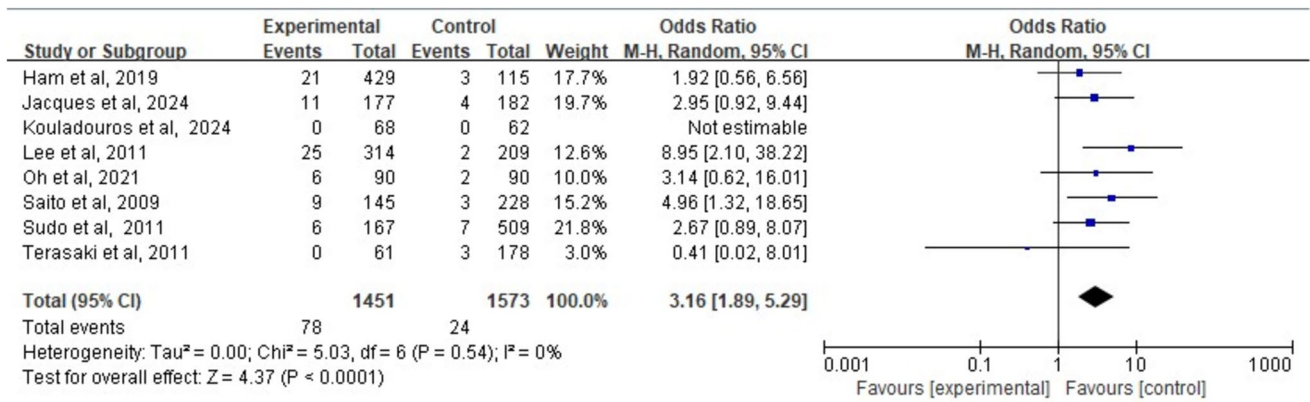


Fig. 8 Forest plot, perforation rate

Fig. 9 Evaluation of publication bias for perforation using funnel plot

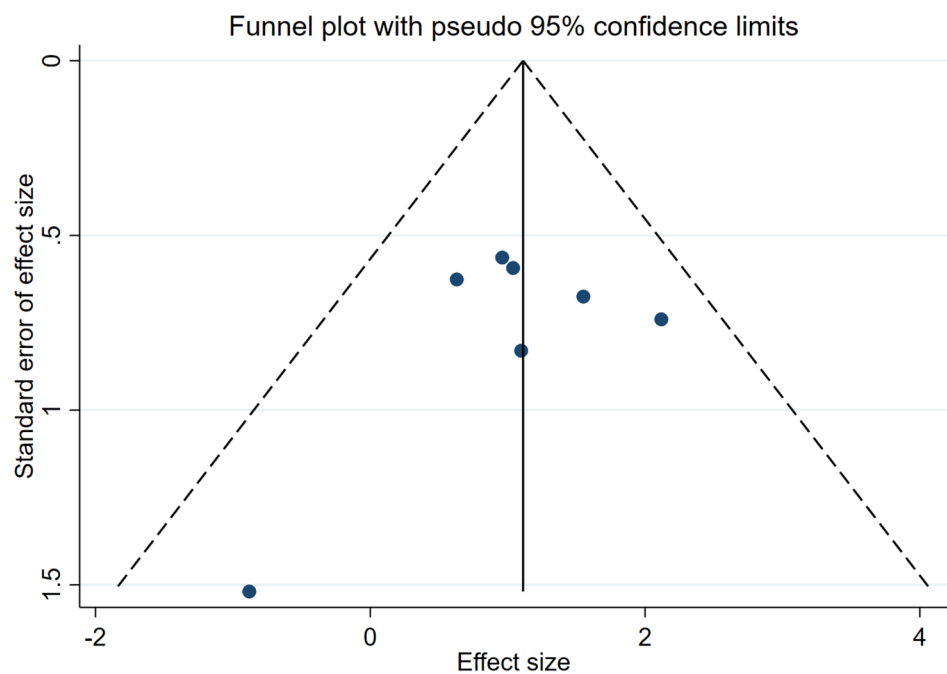
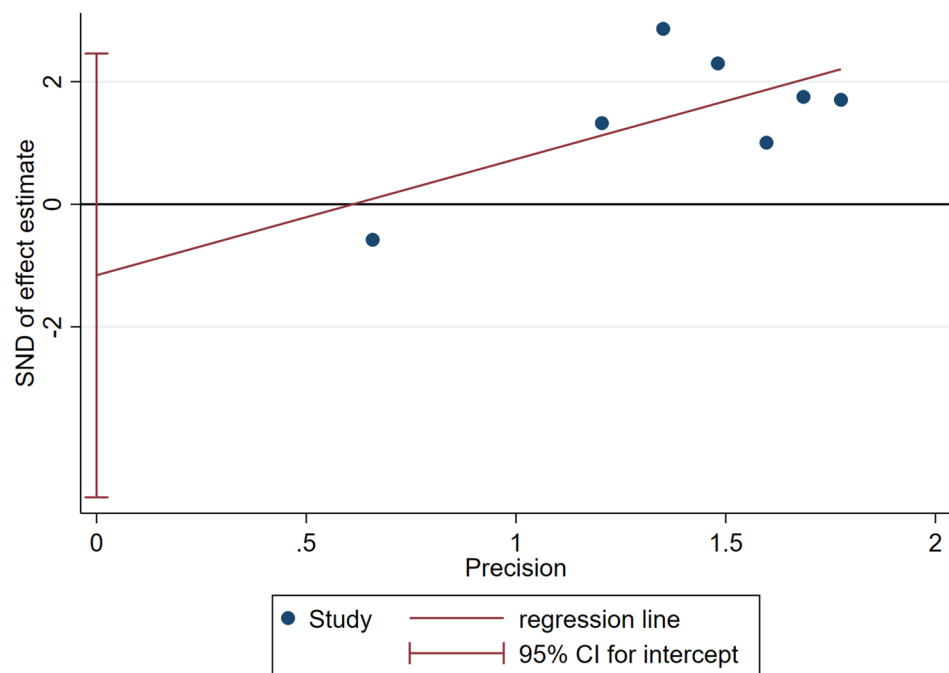


Fig. 10 Evaluation of publication bias for perforation using Egger's test



alternative; however, patients should be fully informed of the higher recurrence risk and subject to more intensive surveillance protocols. Future research should focus on optimizing techniques, shortening the learning curve, and clarifying the optimal treatment strategies for different LST subtypes and locations.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s10151-026-03341-5>.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no competing interests.

Ethical approval and Informed consent This study is a systematic review and meta-analysis based on previously published data; therefore, no new ethical approval or direct patient informed consent was required. All analyzed primary studies were assessed to ensure they had obtained appropriate ethical review board approval and, where applicable, documented patient informed consent. This work was conducted in accordance with the principles of the Declaration of Helsinki and

adhered to the reporting guidelines of the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA 2020).

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